OFFICIAL INCIDENT AUDIT | OPERATIONAL SAFE | ID: CSM-202610231520

Crowd Dynamics & Safety Observational Report

Comprehensive spatio-temporal telemetry, crush risk estimation, and automated incident documentation.

Engine: YOLOv8s + VIVA Observational L1
Generated: 23/10/2026, 15:24:58
Source: mall_dense_crowd.mp4

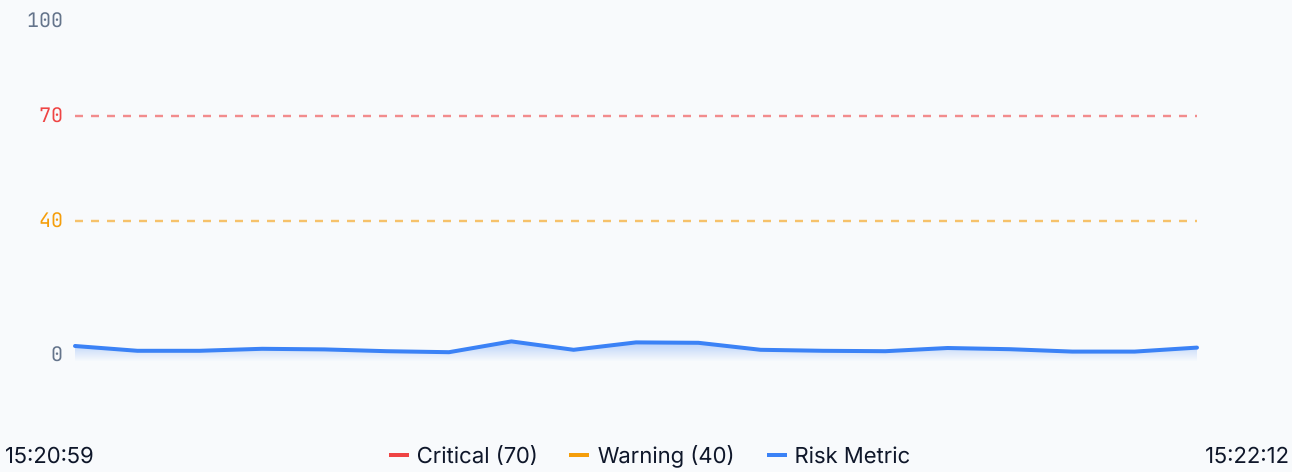
NORMAL OPERATIONAL BASELINE
Crowd flow and spatial density parameters remained safely below alerting thresholds throughout the monitored session.

PEAK RISK SCORE	PEAK HEADCOUNT	PEAK CRUSH / ASPHYXIA
9.6	38	0.0%
Threshold: 70.0 (High)	Max concurrent persons	Compressive load index

AVERAGE DENSITY	CRITICAL ALERTS	MONITORED DURATION
10.0%	0	19s
Spatio-temporal mean	Automated triggers	19s (596 frames)

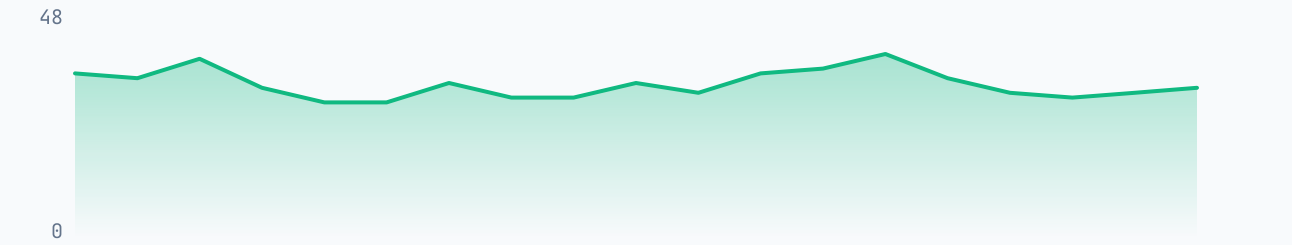
Spatio-Temporal Risk Trajectory (R_z[t])

Historical progression of composite risk score across 1-second sample intervals. Dashed lines indicate Warning (40) and Critical (70) thresholds.



Concurrent Headcount Trajectory (Persons N[t])

Samples: 19 | Peak: 38 persons



Observational Signal Methodology & Verification

Physics-grounded crowd dynamics strictly computed from observational CCTV telemetry without synthetic simulation.

SIGNAL IDENTIFIER	PHYSICAL DIMENSION	MATHEMATICAL FORMULATION	IMPACT ON CRUSH ASSESSMENT
`D_z[n]`	Spatial Density	Bounding box area summation over localized zone grid: `∑ Area(B_i) / Area(Z)`	Direct indicator of crowd packaging. Densities above 60% trigger localized stagnation alerts.
`C_z[n]`	Congestion Factor	Inverse velocity magnitude decay: `1.0 / (||v_mean|| + ε)`	High density coupled with zero velocity indicates severe crowd locking/crush potential.
`T_z[n]`	Residence Duration	Temporal persistence filter over consecutive frames: `α · T[n−1] + (1−α) · Δt`	Distinguishes moving pedestrian streams from dangerous multi-minute bottleneck pileups.
`S_z[n]`	Asphyxia / Compressive Risk	Non-linear product of local density and velocity arrest: `D_z × (1.0 − v_norm)`	Estimates mechanical pressure on thoracic cavities during crowd collapse scenarios.

Captured Incident Evidence Gallery

0 Snapshots Logged

High-risk frames autonomously recorded during critical safety anomalies.

No critical incident snapshots recorded for this session.

Safety Protocol & Action Recommendations

- Continuous Standard Protocol: Standard baseline monitoring active.
- Hardware Health: All telemetry and ByteTrack tracking pipelines operating with normal confidence bounds.

SURVEILLANCE OFFICER

Automated Vision Engine V3

Signature / System Auth

OPERATIONS SUPERVISOR

Verification Pending

Authorized Sign-Off

COMPLIANCE STATUS

PASSED — SAFE LIMITS

ISO / Safety Protocol 2026